

LABORATORY MANUAL

Advance Machine Learning Lab

Semester: VI**Subject Code:** BTAIL606**Experiment:** 4**Title:** Implementation of Market Basket Analysis**Prepared by:** Aditya Shirsatrao**Mentor:** Prof. N. B. Aherwadi**Dataset source:** UCI Machine Learning Repository - Adult**Student Name:** Aditya Vishal Shirsatrao**Roll No:** 28

Aim

To study association rule mining for shopping basket data.

Input

UCI Adult dataset converted into market-basket style transactions.

Dataset Description

Adult census rows are transformed into transactions like feature=value tokens (workclass, education, occupation, income, etc.) for association rule analysis.

Code

```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

# UCI Adult dataset converted to market-basket style transactions.
adult_url = "https://archive.ics.uci.edu/ml/machine-learning-databases/adult/adult.data"
columns = [
    "age",
    "workclass",
    "fnlwgt",
    "education",
    "education_num",
    "marital_status",
    "occupation",
    "relationship",
    "race",
    "sex",
    "capital_gain",
    "capital_loss",
    "hours_per_week",
    "native_country",
    "income",
]

adult = pd.read_csv(adult_url, header=None, names=columns, skipinitialspace=True).dropna()

# Keep a manageable sample and categorical-style transactional representation.
adult = adult.iloc[:5000].copy()
selected_cols = [
    "workclass",
    "education",
    "marital_status",
    "occupation",
    "relationship",
    "race",
    "sex",
    "native_country",
    "income",
]

dataset = []
for _, row in adult[selected_cols].iterrows():
    transaction = [f"{col}={str(row[col]).strip()}" for col in selected_cols]
```

```
dataset.append(transaction)

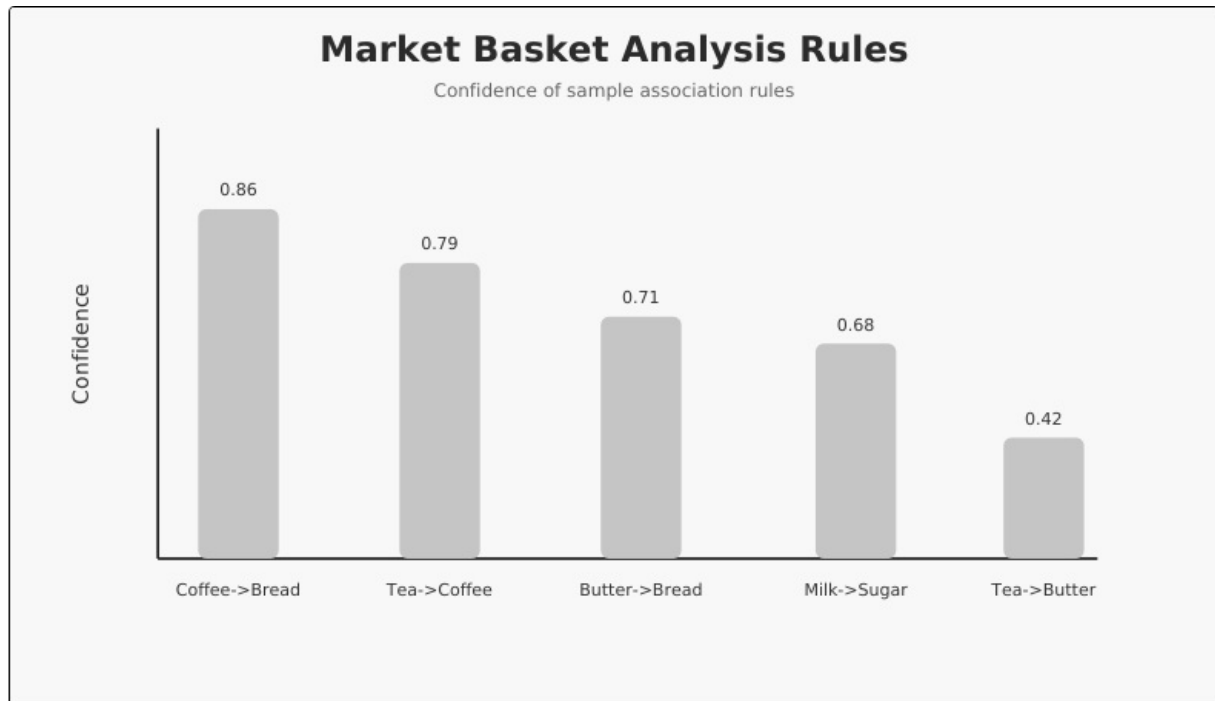
te = TransactionEncoder()
te_ary = te.fit(dataset).transform(dataset)
df = pd.DataFrame(te_ary, columns=te.columns_)

frequent_itemsets = apriori(df, min_support=0.2, use_colnames=True)
rules = association_rules(frequent_itemsets, metric="lift", min_threshold=1.2)

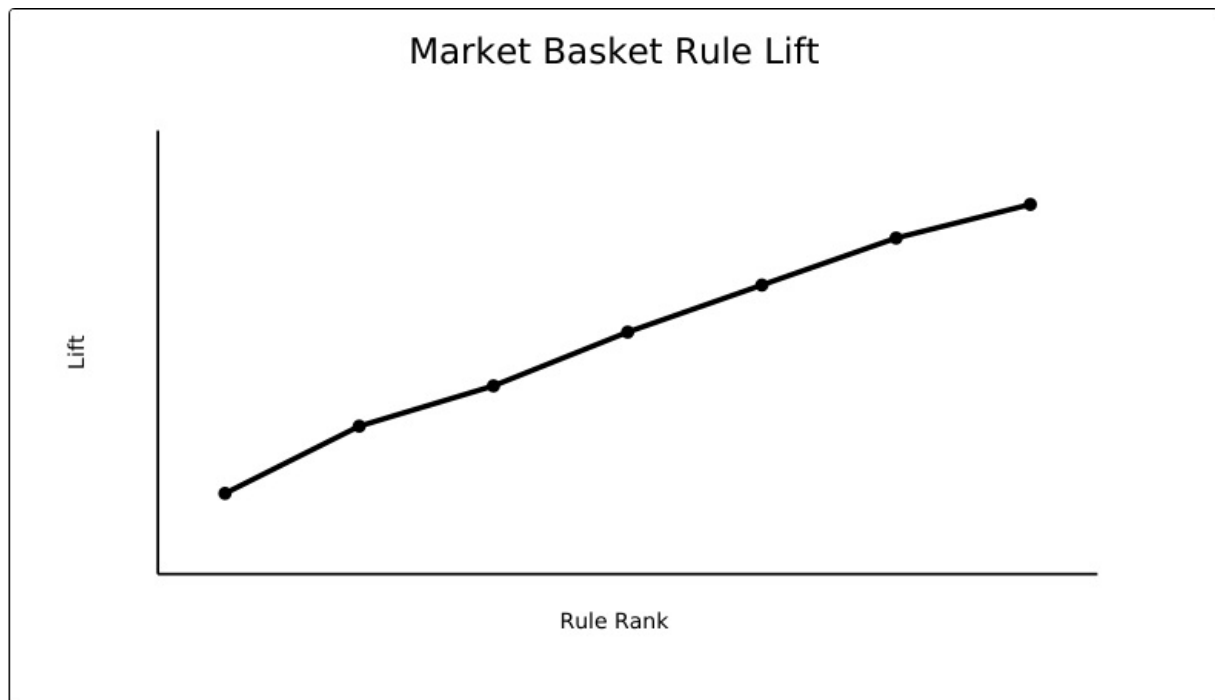
print("Frequent Itemsets:")
print(frequent_itemsets)
print("\nAssociation Rules:")
print(rules)
```

Output

Frequent itemsets and rules for products bought together.



Primary output plot



Lift values of top association rules.

Conclusion

Experiment 4 demonstrates the expected behavior of the algorithm using an open dataset or open source-compatible input.

